

T3: physical symmetric Weinberg coefficient C_5

The physical coefficient is displayed without the redundant ordered Matchete column. For all 16 models the saved ordered coefficients have the same mass-dependent structure after removing the flavour tensor and exact class prefactor:

$$C_{5,m}^{ij} = \kappa_m \mathcal{F}_{\text{match}}(M_F, m_1, m_2, \bar{\mu}, \epsilon) (\overline{y_{1,i,r}} \overline{y_{2,j,r}} + \overline{y_{1,j,r}} \overline{y_{2,i,r}}).$$

The repeated fermion index r is summed as in the saved output. The common factor $\mathcal{F}_{\text{match}}$ includes λ_5 , $\hbar$, the Matchete integral I_3 , and its original mass and regulator dependence. It is defined exactly by the common kernel in `C5_common_kernel.txt`; no loop-integral simplification or renormalisation has been assumed here.

Model	Class	(d_{S_1}, d_{S_2}, d_F)	κ_m
T3-A-m4	A	(1, 3, 2)	$\frac{2\sqrt{3}}{3}$
T3-A-m2	A	(1, 3, 2)	$\frac{2\sqrt{3}}{3}$
T3-A-p0	A	(1, 3, 2)	$\frac{2\sqrt{3}}{3}$
T3-B-m3	B	(2, 2, 1)	$\frac{2\sqrt{3}}{3}$
T3-B-m1	B	(2, 2, 1)	$\frac{2\sqrt{3}}{3}$
T3-B-p1	B	(2, 2, 1)	$\frac{2\sqrt{3}}{3}$
T3-C-m3	C	(2, 2, 3)	$-\frac{2}{3}$
T3-C-m1	C	(2, 2, 3)	$-\frac{2}{3}$
T3-C-p1	C	(2, 2, 3)	$-\frac{2}{3}$
T3-D-m2	D	(3, 1, 2)	$\frac{2\sqrt{3}}{3}$
T3-D-p0	D	(3, 1, 2)	$\frac{2\sqrt{3}}{3}$
T3-D-p2	D	(3, 1, 2)	$\frac{2\sqrt{3}}{3}$
T3-E-m4	E	(3, 3, 2)	$-\frac{2\sqrt{6}}{3}$
T3-E-m2	E	(3, 3, 2)	$-\frac{2\sqrt{6}}{3}$
T3-E-p0	E	(3, 3, 2)	$-\frac{2\sqrt{6}}{3}$
T3-E-p2	E	(3, 3, 2)	$-\frac{2\sqrt{6}}{3}$

Scope. The prefactors and common kernel are obtained by literal factor extraction from the previously generated Matchete expressions, not from an independent matching calculation. The full original expression remains preserved separately.